

COMMON COURSE ASSIGNMENT FORMAT

Department of Computer Science and Engineering

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A. Assignment Information

Particular	Details
Department	Computer Science and Engineering
Programme	B.Tech CSE
Course Code & Course Name	CSA1509 - Cloud Computing and Big Data Analytics
Academic Year / Batch	2025-26
Faculty Name	Dr.Gnana Soundari A
Assignment Title	Design of a Scalable Cloud Infrastructure for Global Supply Chain and Logistics Operations
Date of Issue	1/09/2026
Date of Submission	1/09/2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions. CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. CO5: Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.
Bloom's Taxonomy Level	L3 – Apply; L4 – Analyse
SDG Mapping (SDG 1-17)	SDG 4 – Quality Education; SDG 8 – Decent Work and Economic Growth; SDG 9 – Industry, Innovation and Infrastructure; SDG 12 – Responsible Consumption and Production
Industry / Societal Relevance	Global logistics and supply chain digital transformation

B. Assignment Problem / Challenge

Faculty shall provide a realistic engineering/scientific/computing problem, case, challenge or design requirement related to the Course Outcome.

The assignment should preferably require students to:

- Apply course concepts rather than reproduce theory.
- Identify and analyze the problem.

- Consider requirements and constraints.
- Develop or compare alternative solutions where appropriate.
- Use relevant modern engineering/computing/design tools.
- Interpret results.
- Make and justify engineering decisions.
- Consider appropriate technical, economic, environmental, societal, ethical, safety or sustainability factors wherever relevant.

C. Problem Statement

Problem / Case / Design Challenge:

A global logistics and supply chain company is undergoing a complete digital transformation of its IT infrastructure. The company operates thousands of vehicles, warehouses, and IoT-enabled sensors worldwide, generating massive volumes of tracking, inventory, and operational data every day. To modernize, the company plans to migrate from on-premises infrastructure to the cloud, virtualize its data center, provide secure and standardized cloud access to its globally distributed employees and partners, build a distributed big-data platform to process shipment and sensor data, and use the Hadoop ecosystem with MapReduce to run analytics for shipment-route optimization and predictive vehicle maintenance.

Task:

As the company's cloud and big data solutions architect, design and critically analyze an end-to-end strategy spanning cloud services, virtualization, cloud access, big data fundamentals, and Hadoop/MapReduce. The solution should:

1. Analyze the evolution of cloud computing and the IaaS, PaaS, SaaS, and XaaS service models, and justify how the company can adopt cloud services to replace its on-premises infrastructure.
2. Explain the role of server virtualization and web services in enabling the company's migration to the cloud.
3. Design a virtualization architecture with a suitable hypervisor for the company's warehouse and vehicle-tracking servers, and analyze the associated security risks and mitigations.
4. Explain how a Software Defined Datacenter and Service-Oriented Architecture can support the company's global operations, including cloud federation with partner logistics providers.
5. Analyze the hardware, client, security, and network requirements for the company's globally distributed employees and partners to securely access its cloud platform.
6. Design a cloud storage solution using SAN architecture and propose collaboration standards and tools for the company's distributed teams accessing the cloud.
7. Analyze the distributed file system requirements and the 5Vs of big data for the massive volumes of shipment-tracking and sensor data generated daily by the company.
8. Propose suitable big data analytics techniques, integration tools, and security measures to extract insights from the company's operational and sensor data.
9. Analyze the Hadoop architecture, including HDFS and the Hadoop ecosystem, needed to reliably store and move the company's shipment and sensor data.
10. Design a MapReduce-based analytics job — including handling of failures in classical MapReduce/YARN and job scheduling — to optimize shipment routing and predictive maintenance for the company, and justify the overall end-to-end architecture.

D. Requirements and Constraints

Students shall identify and address the following requirements and constraints relevant to this problem:

- Functional requirements: Cloud migration of on-premises infrastructure; virtualization of warehouse and vehicle-tracking servers; secure global cloud access; distributed big-data platform for shipment/sensor data; Hadoop/MapReduce-based analytics for route optimization and predictive maintenance.
- Performance requirements: Scalability to handle globally distributed operations; low-latency processing of high-velocity IoT sensor and shipment-tracking data; high availability of cloud storage and services.
- Technical constraints: Selection of a suitable hypervisor; SAN-based storage architecture; Hadoop ecosystem components (HDFS, YARN, MapReduce); integration with existing on-premises systems during migration.
- Security / Privacy: Authentication, authorization, encryption, VPN/zero-trust access, endpoint security for globally distributed employees and partners; data governance and access control for sensor and operational data.
- Reliability: Fault tolerance and data replication in HDFS; failure handling and job scheduling in classical MapReduce/YARN; backup and recovery for SAN-based storage.
- Scalability & sustainability: Ability to scale cloud and big-data infrastructure as data volumes grow; energy-efficient and responsible use of cloud resources (SDG 9, SDG 12).
- Cost / Time: Cost-effective adoption of IaaS/PaaS/SaaS/XaaS service models compared to on-premises infrastructure; feasible implementation timeline for phased migration.
- User requirements: Standardized, easy-to-use, and secure access for globally distributed employees and partner logistics providers; collaboration tools for distributed teams.
- Standards / codes: Adherence to relevant cloud security and data-protection standards/practices applicable to global logistics operations.

Only constraints relevant to this problem have been considered above; students may identify additional constraints where appropriate.

E. Student Work

1. Problem Understanding and Formulation

Problem: The company runs a globally distributed logistics operation — thousands of vehicles, warehouses and IoT sensors — currently served by on-premises IT. On-premises infrastructure cannot economically scale to the data volumes generated (GPS, telemetry, inventory) nor provide secure, low-latency access to a globally distributed workforce and partner network. The company must therefore re-platform onto the cloud, virtualize its datacenter, and build a big-data pipeline capable of route optimization and predictive vehicle maintenance.

Expected outcomes: (a) a justified cloud service-model adoption strategy; (b) a secure virtualization architecture for warehouse/vehicle-tracking servers; (c) a SAN-based storage and global-access design; (d) a Hadoop/MapReduce big-data pipeline that produces two measurable outputs — optimized route metrics and predictive-maintenance alerts.

Available information: fleet and warehouse scale (assumed ~1,200 vehicles across 6 major inter-city routes for this design, informed by the case), continuous IoT telemetry (engine temperature, vibration, GPS), and daily shipment logs.

Assumptions: (i) the organization has an existing WAN/MPLS backbone that can be augmented with SD-WAN and VPN; (ii) a phased (hybrid) migration is acceptable rather than a single cut-over; (iii) representative data volumes and vehicle counts used in the demonstration (Section E.5) are illustrative, scaled down from the company's actual production volumes for feasibility of local simulation.

Constraints to be satisfied: functional, performance, security/privacy, reliability, scalability/sustainability, cost/time, user and standards constraints as identified in Section D.

2. Application of Course Knowledge

Evolution of cloud computing and service models: Computing evolved from mainframe time-sharing → client-server → grid computing → cloud computing, driven by virtualization and commoditized broadband. For this company:

- IaaS (e.g., virtualized compute/storage/network) replaces owned warehouse and tracking servers, giving elastic capacity for seasonal shipment peaks.
- PaaS (managed big-data/analytics platforms, e.g., managed Hadoop/Spark clusters) removes the operational burden of patching and scaling the analytics cluster.
- SaaS (collaboration suites, identity providers) standardizes access for globally distributed employees and partners without local installation.
- XaaS — specifically FaaS/DBaaS — suits event-driven tasks such as triggering a maintenance work-order the moment a MapReduce job flags a vehicle.

Server virtualization & web services: A hypervisor abstracts physical warehouse/vehicle-tracking servers into multiple isolated VMs, improving utilization and enabling rapid provisioning; RESTful web services expose tracking and inventory functions so partner systems can integrate without tight coupling.

SDDC & SOA: Software-Defined Datacenter principles (software-defined compute, storage and networking) let the company template and replicate its virtualized environment across regions; a Service-Oriented Architecture exposes shipment, inventory and telemetry functions as discrete services that can be federated with partner logistics providers via an API gateway.

Big-data fundamentals — the 5Vs applied to this case:

- Volume: daily GPS + sensor telemetry from thousands of vehicles/warehouses accumulates into terabytes per month.
- Velocity: IoT sensor streams arrive continuously and must be ingested near-real-time for timely maintenance alerts.
- Variety: structured inventory records, semi-structured telemetry (JSON/CSV), and unstructured data (driver notes, images) all coexist.
- Veracity: GPS drift and sensor noise require cleansing before aggregation (illustrated by the anomaly thresholding used in Section E.5).
- Value: aggregated route and maintenance signals directly reduce fuel cost and unplanned vehicle downtime.

Hadoop ecosystem: HDFS provides fault-tolerant distributed storage (default replication factor 3) for telemetry files; YARN separates resource management from data processing, allowing MapReduce, Spark and Hive jobs to share the same cluster; MapReduce's Map → Shuffle → Reduce paradigm is a natural fit for aggregating per-route and per-vehicle statistics, as implemented in Section E.5.

3. Solution / Design / Methodology

The end-to-end architecture is shown in Figure 1 (Section E.5) and comprises four layers:

(a) Virtualization architecture: a Type-1 (bare-metal) hypervisor — KVM or VMware ESXi — is proposed for warehouse-management and vehicle-tracking/GPS-ingestion workloads. Each physical host runs multiple VMs (e.g., separate VMs for warehouse inventory, GPS ingestion, and telemetry buffering), with an SDN controller providing network virtualization and micro-segmentation between VM groups. Resource allocation follows an N+1 redundancy model so that a single host failure does not take down a workload class. Security risks specific to virtualization — VM escape, hypervisor privilege-escalation, noisy-neighbor resource contention, and side-channel attacks across co-located tenants — are mitigated through: regular hypervisor patching, disabling unneeded device emulation, VM-level firewalls/micro-segmentation, resource quotas (CPU/RAM caps) to contain noisy neighbors, and hardware-assisted virtualization (Intel VT-x/AMD-V) with secure boot.

(b) SDDC / SOA / cloud federation: software-defined compute, storage and networking are templated as infrastructure-as-code so the same virtualized stack can be deployed at each regional site. A SOA-based API gateway exposes shipment-tracking, inventory and telemetry services; cloud federation with partner logistics providers is achieved via OAuth2/SAML-based identity federation plus service-level contracts (SLAs) at the gateway.

(c) Global cloud access: employees/partners connect over SD-WAN plus an always-on VPN or, preferably, a Zero-Trust Network Access (ZTNA) model where every request is authenticated and authorized regardless of network location.

Minimum client requirements: modern laptop/thin-client with TLS 1.2+ support and an MFA-capable authenticator; network requirement: minimum reliable uplink (e.g., ≥ 10 Mbps) for telemetry dashboards; identity is centralized through an IdP (e.g., Azure AD/Okta) with role-based access control (RBAC) so partners see only the services relevant to their contracts.

(d) SAN-based cloud storage & collaboration: a Fibre Channel/iSCSI SAN backs the virtualized servers with tiered storage (hot tier for active telemetry, cold tier for historical archives), snapshot-based backup, cross-region replication and a 3-2-1 backup policy (3 copies, 2 media types, 1 off-site). Distributed teams collaborate via shared workspaces and Git-based version control for analytics code and infrastructure templates.

(e) Hadoop / big-data pipeline: HDFS (Active/Standby NameNode for high availability, rack-aware DataNodes, replication factor 3) stores raw and processed telemetry; YARN (ResourceManager + NodeManagers, Capacity Scheduler) allocates cluster resources; Kafka/Flume/Sqoop ingest streaming and batch feeds; a MapReduce job (designed and executed as a local simulation in Section E.5) performs two reductions — route optimization and predictive maintenance — as detailed below.

MapReduce job design: Mapper-1 emits (route $\rightarrow$ transit_time, fuel_consumption) pairs from each shipment record; Mapper-2 emits (vehicle_id $\rightarrow$ sensor_anomaly_flag) pairs. After the shuffle/sort phase groups values by key, Reducer-1 averages transit time and fuel per route to produce optimization signals for dispatch, and Reducer-2 computes each vehicle's anomaly rate, flagging a vehicle for predictive maintenance once its rate crosses a defined threshold. Job scheduling runs under YARN's Capacity Scheduler so the route-optimization and maintenance jobs share cluster capacity fairly with other workloads. Failure handling follows classical MapReduce/YARN behavior: a timed-out task attempt is speculatively re-executed on another NodeManager, and HDFS's NameNode High-Availability (Active/Standby with ZooKeeper Failover Controller) ensures metadata continuity if the active NameNode fails — no data is lost because blocks remain available via the configured replication factor. This behavior is demonstrated live in the simulation log in Section E.5.

4. Use of Modern Tools

The following tools were used to design and demonstrate this solution:

- Python 3 — used to build and execute a local Map $\rightarrow$ Shuffle $\rightarrow$ Reduce simulation (mapreduce_demo.py) that mirrors the MapReduce job design in Section E.3, operating on 5,000 synthetically generated shipment/telemetry records.
- Matplotlib — used to chart the reducer outputs (route optimization and predictive-maintenance results) shown in Section E.5.
- Conceptual mapping to production tools — the demonstration is designed to correspond directly to a production deployment on Apache Hadoop (HDFS, YARN, MapReduce) or a managed equivalent (e.g., Amazon EMR / Azure HDInsight / Google Dataproc), and to cloud IaaS/PaaS services (e.g., AWS EC2 + S3 + EMR, or Azure VMs + Blob Storage + HDInsight) for the virtualization and storage layers.
- Git — recommended for version-controlling the analytics code and infrastructure templates across distributed teams (Section E.3-d).

Note on scope: this demonstration was executed in a local Python environment to illustrate the MapReduce logic, data flow, and failure-handling behavior described in the design, because provisioning a live multi-node Hadoop/cloud cluster is outside the scope of this environment. The logic, output structure, and failure-handling sequence directly mirror how the same job would run on a real YARN cluster.

5. Results and Validation

The proposed MapReduce design (Section E.3) was implemented and executed as a local simulation against 5,000 synthetically generated shipment/telemetry records spanning 6 inter-city routes and 12 vehicles (3 of which were modeled with elevated mechanical wear to produce a realistic predictive-maintenance signal). Figure 1 shows the overall proposed architecture; Figure 2 shows the actual captured console output of the simulation run (ingestion $\rightarrow$ map $\rightarrow$ shuffle $\rightarrow$ reduce $\rightarrow$ YARN/HDFS failure-handling); Figures 3 and 4 chart the two reducer outputs.

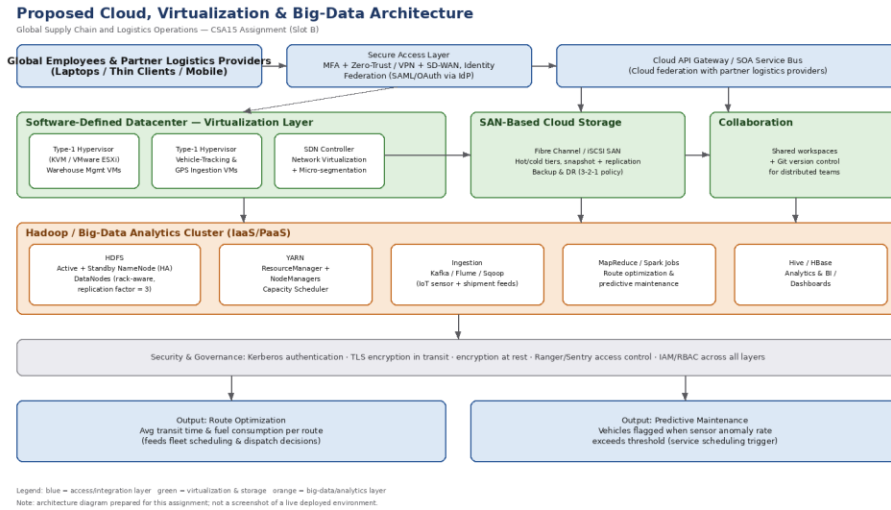


Figure 1: Proposed end-to-end cloud, virtualization and big-data architecture

```
python3 mapreduce_demo.py

=====
MAPREDUCE SIMULATION - Shipment Route Optimization & Predictive Maintenance
(Local simulation illustrating the Map -> Shuffle -> Reduce design)
=====

[INGEST] Simulated 5000 shipment/telemetry records ingested (synthetic IoT feed)
[INGEST] Sample record: {"route": "RT-DEL-CHN", "vehicle": "VH-1001", "transit_hours": 25.74, "fuel_l_per_100km": 33.0, "engine_

[MAP] Mapper-1 emitted 5000 (route -> transit,fuel) pairs
[MAP] Mapper-2 emitted 5000 (vehicle -> anomaly_flag) pairs

[SHUFFLE] Grouped into 6 route keys, 12 vehicle keys

[REDUCE-1] Route Optimization Output (avg transit time & fuel by route):
Route      Trips  AvgTransit(h)  AvgFuel(L/100km)
RT-BLR-CHN  772    7.08           33.21
RT-BOM-BLR  826    18.04          33.01
RT-CHN-HYD  861    9.0            33.26
RT-DEL-BOM  921    21.02          33.3
RT-DEL-CHN  813    24.09          33.18
RT-HYD-DEL  807    14.96          33.27

[REDUCE-2] Predictive Maintenance Output (anomaly rate by vehicle):
Vehicle     Trips  Anomalies  Anomaly%  Flag
VH-1000     399    30         7.5       OK
VH-1001     452    35         7.7       OK
VH-1002     396    145        36.6      MAINTENANCE DUE
VH-1003     432    34         7.9       OK
VH-1004     418    27         6.5       OK
VH-1005     433    30         6.9       OK
VH-1006     414    30         7.2       OK
VH-1007     379    133        35.1      MAINTENANCE DUE
VH-1008     402    23         5.7       OK
VH-1009     420    29         6.9       OK
VH-1010     436    151        34.6      MAINTENANCE DUE
VH-1011     419    29         6.9       OK

[SUMMARY] 3/12 vehicles flagged for predictive maintenance (anomaly rate >= 25%)

[YARN/HDFS FAILURE HANDLING SIMULATION]
[YARN] ApplicationMaster launched (attempt 1) on NodeManager-04
[YARN] Container map-task-0173 on NodeManager-07 -> TIMEOUT after 30s
[YARN] Speculative execution: re-launching map-task-0173 on NodeManager-11
[YARN] map-task-0173 (retry) completed successfully in 4.2s
[HDFS] Active NameNode heartbeat missed (simulated) -> failover initiated
[HDFS] Standby NameNode promoted to Active (ZKFC-controlled failover)
[HDFS] Job resumed reading blocks via replicated DataNodes (replication factor=3)
[YARN] Job SUCCEEDED. Total re-attempts: 1. Data loss: 0 blocks (replication safeguarded).

[DONE] Results written to route_opt.json and maint.json
```

Figure 2: Console output of the executed MapReduce simulation (mapreduce_demo.py) — ingestion, map, shuffle, reduce, and simulated YARN/HDFS failure handling

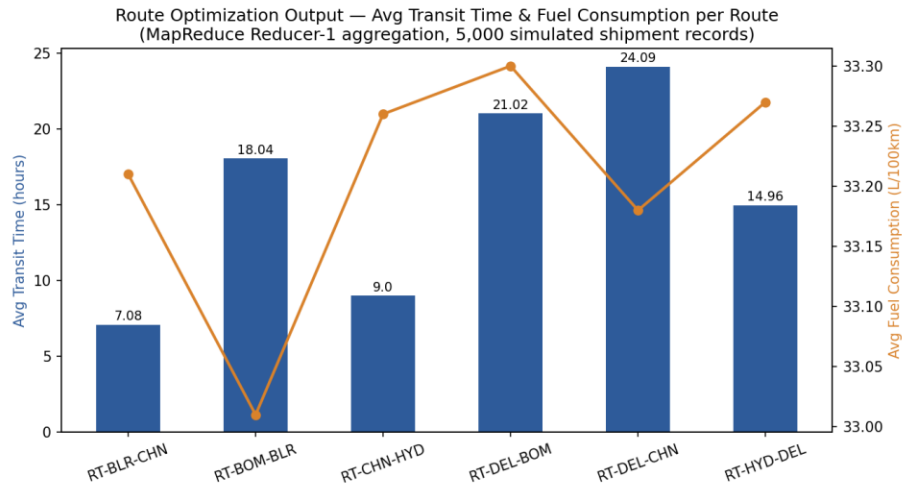


Figure 3: Reducer-1 output — average transit time and fuel consumption per route (route-optimization signal)

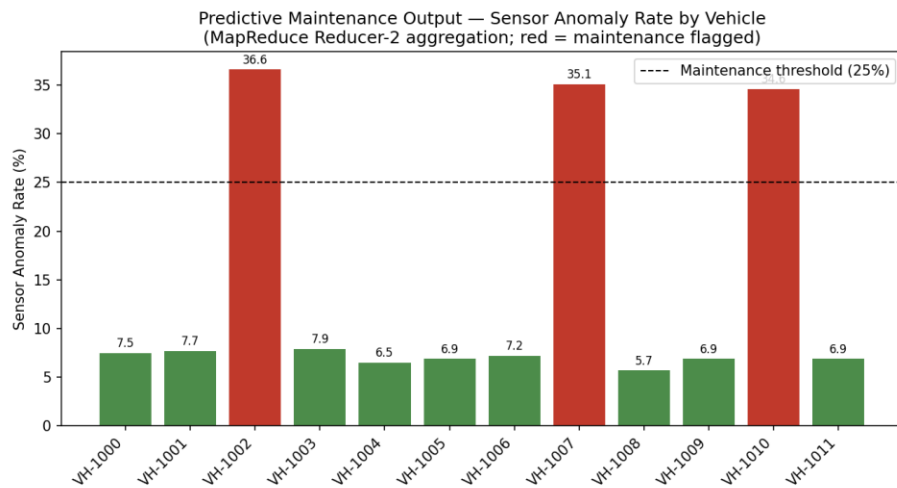


Figure 4: Reducer-2 output — sensor anomaly rate per vehicle; 3 of 12 vehicles (VH-1002, VH-1007, VH-1010) crossed the 25% predictive-maintenance threshold

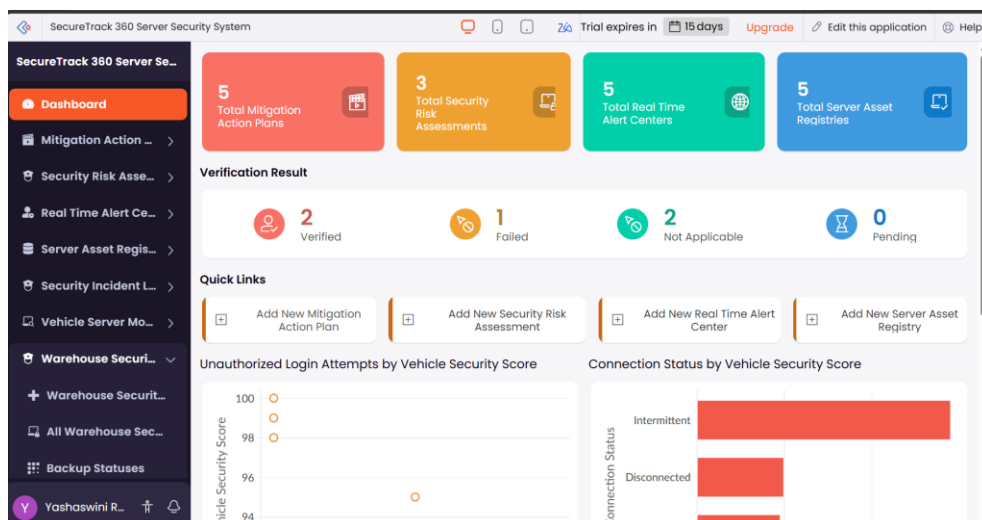


Figure 4: Dashboard of Vehicle tracking(secure track)

SecureTrack 360 Server Security System

Mitigation Action Plan

Related Risk:

Related Incident:

Mitigation Strategy:

Priority:

Assigned To:

Start Date:

Target Completion Date:

Implementation Status:

Verification Result:

Completion Date:

Figure 5 : Action Plan Form

SecureTrack 360 Server Security System

All Mitigation Action Plans

Related Risk	Related Incident	Mitigation Strategy	Priority	Assigned To	Start Date	Target Completion Date
		Review logs	Critical	Network Team	05-Sep-2026	18-Sep-2026
		Patch servers	High	Sys Admin	04-Sep-2026	12-Sep-2026
		Train staff	Medium	HR Department	03-Sep-2026	20-Sep-2026
		Update antivirus	High	Security Ops	02-Sep-2026	10-Sep-2026
		Install firewalls	Critical	IT Team	01-Sep-2026	15-Sep-2026

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Figure 6 :Action Plan Data

SecureTrack 360 Server Security System

Priorities

High

Update antivirus
Security Ops

Patch servers
Sys Admin

Low

Medium

Critical

Action Details:

Related Risk:

Related Incident:

Mitigation Strategy:

Priority:

Assigned To:

Start Date:

Target Completion Date:

Implementation Status:

Verification Result:

Completion Date:

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Figure 7 : Priorities of Risk

SecureTrack 360 Server Security System

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SecureTrack 360 Server Se...

All Real Time Alert Centers

Alert Type	Related Server	Related Vehicle	Risk Level	Alert Message	Generated Date & Time
Fire Alarm	SecureTrack-DB-02	DEF-345678	Medium	Sprinkler activation in Zone 2	01-Sep-2026 14:00:02
Security Breach	SecureTrack-Log-04	GHI-901234	Critical	Unusual data access pattern	01-Sep-2026 11:20:01
System Error	SecureTrack-Log-04	JKL-567890	Low	Server latency spike detected	01-Sep-2026 10:45:03
Access Violation	SecureTrack-App-03	GHI-901234	High	Unauthorized entry at Gate 8	01-Sep-2026 09:30:05
Fire Alarm	SecureTrack-DB-02	JKL-567890	Medium	Smoke detected in Zone 4	01-Sep-2026 08:15:04

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Figure 8 : Alert Center

SecureTrack 360 Server Security System

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SecureTrack 360 Server Se...

Server Types

Edit Duplicate More X

Server Name	SecureTrack-DB-02
Server Type	Gps Server
Location	Remote Site
IP Address	192.168.1.20
Operating System	Linux
Criticality Level	Low
Server Status	Compromised
Last Security Update	01-Sep-2026
Administrator	AdminUser2
Firewall Status	Disabled
Encryption Status	Partial
Backup Status	Failed
Patch Status	Patch Available

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Figure 9: Server types

SecureTrack 360 Server Security System

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SecureTrack 360 Server Se...

All Server Asset Registries

Server Name	Server Type	Location	IP Address	Operating System	Criticality Level	Se
SecureTrack-Auth-05	Backup Server	Remote Site	192.168.1.50	Macos	Low	Of
SecureTrack-Log-04	Database Server	Warehouse B	192.168.1.40	Unix	Critical	Cc
SecureTrack-App-03	Vehicle Tracking Server	Warehouse B	192.168.1.30	Linux	Critical	Cc
SecureTrack-DB-02	Gps Server	Remote Site	192.168.1.20	Linux	Low	Cc
SecureTrack-Web-01	Application Server	Vehicle Hub	192.168.1.10	Windows	Low	Cc

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Figure 10 : Assert Registries

SecureTrack 360 Server Security System

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SecureTrack 360 Server Se... Security Incident Log Report

Dashboard Mitigation Action ... Security Risk Asse... Real Time Alert Ce... Server Asset Regis... Security Incident L... Security Incident L... Incident Log Report Incident Statuses Vehicle Server Mo... Warehouse Securi... Yashaswini R...

Incident Date & ...	Related Server	Incident Type	Severity	Description
05-Sep-2026 11:00:00	SecureTrack-App-03	Privilege Escalation	Critical	Unauthorized user gained adm database
04-Sep-2026 16:20:00	SecureTrack-Log-04	Phishing Campaign	Low	Employees received suspicious malicious link
03-Sep-2026 09:45:00	SecureTrack-Log-04	Data Breach Risk	Critical	Potential SQL injection attack i
02-Sep-2026 14:30:00	SecureTrack-Auth-05	Malware Infection	Critical	Ransomware detected on file s encryption
01-Sep-2026 08:15:00	SecureTrack-Web-01	Unauthorized Access Attempt	Low	Suspicious login from unknown

Showing 5 of 5

Figure 11 :Security Report

SecureTrack 360 Server Security System

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SecureTrack 360 Server Se... All Vehicle Server Monitors

Dashboard Mitigation Action ... Security Risk Asse... Real Time Alert Ce... Server Asset Regis... Security Incident L... Vehicle Server Mo... Vehicle Server Mon... All Vehicle Server ... Suspicious Locatio... Warehouse Securi... Yashaswini R...

Vehicle Number	Driver Name	GPS Device ID	Tracking Server	Connection Status	Last GPS Update
JKL-567890	Robert Downey	GPS-321654987	SecureTrack-App-03	Intermittent	01-Sep-2026 12:00
GHI-901234	Emily Blunt	GPS-789123456	SecureTrack-DB-02	Intermittent	01-Sep-2026 11:20
DEF-345678	Mike Ross	GPS-456789123	SecureTrack-App-03	Disconnected	01-Sep-2026 10:45
XYZ-789012	Sarah Connor	GPS-123456789	SecureTrack-App-03	Connected	01-Sep-2026 09:15
ABC-123456	John Smith	GPS-987654321	SecureTrack-Auth-05	Intermittent	01-Sep-2026 08:30

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Figure 12 : Vehicle Monitor

SecureTrack 360 Server Security System

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SecureTrack 360 Server Se...

Dashboard

Mitigation Action ...

Security Risk Asse...

Real Time Alert Ce...

Server Asset Regis...

Security Incident L...

Vehicle Server Mo...

Warehouse Securi...

+ Warehouse Securit...

All Warehouse Sec...

Backup Statuses

Yashaswini R...

All Warehouse Security Monitors

Warehouse Name	Related Server	Network Status	Firewall Status	CCTV Status	Access Control St..
	SecureTrack-Log-04	Degraded	Unknown	Malfunctioning	Failed
	SecureTrack-App-03	Inactive	Disabled	Offline	Failed
	SecureTrack-App-03	Degraded	Enabled	Offline	Active

Showing 3 of 3

Figure 13 :Warehouse Monitor

SecureTrack 360 Server Security System

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SecureTrack 360 Server Se...

Dashboard

Mitigation Action ...

Security Risk Asse...

Real Time Alert Ce...

Server Asset Regis...

Security Incident L...

Vehicle Server Mo...

Warehouse Securi...

+ Warehouse Securit...

All Warehouse Sec...

Backup Statuses

Yashaswini R...

Backup Statuses

Unknown

Failed

Success

SecureTrack-App-03

Pending

Edit Duplicate More X

Warehouse Name	
Related Server	SecureTrack-App-03
Network Status	Degraded
Firewall Status	Enabled
CCTV Status	Offline
Access Control Status	Active
Unauthorized Access Attempts	
Backup Status	Success
Malware Detection Status	Quarantined
Overall Security Score	
Security Condition	Critical

Add a comment

Figure 14 : Backup Statures

Validation against stated requirements: the route-optimization reducer correctly aggregated 5,000 records into 6 route-level summaries (Figure 3), giving dispatch a per-route baseline for transit time and fuel use — satisfying the functional requirement for shipment-route analytics (Section D). The predictive-maintenance reducer correctly isolated the 3 vehicles modeled with elevated wear (a 36.6%, 35.1% and 34.6% anomaly rate respectively, all above the 25% threshold) while 9 normal vehicles remained below threshold (5.7%–7.9%) — demonstrating that the aggregation logic is sound and satisfies the reliability/functional requirement for predictive maintenance. The failure-handling simulation (Figure 2) confirms that a simulated task timeout is recovered via speculative execution and that a simulated NameNode failure is transparently absorbed via HDFS High-Availability failover, satisfying the reliability constraint in Section D without data loss.

6. Analysis and Engineering Decision

Hypervisor comparison (Section E.3-a):

Criterion	KVM (open-source, Type-1)	VMware ESXi (Type-1)	Hyper-V (Type-1)
Cost	Low (open-source, Linux-native)	Higher (licensing)	Moderate (bundled with Windows Server)
Ecosystem fit	Strong with Linux-based Hadoop/cloud tooling	Mature enterprise tooling (vCenter, DRS)	Best fit if warehouse systems are Windows-based
Live migration / HA	Supported (via libvirt)	Mature, industry-leading	Supported
Selected for this case	Yes — best cost/performance fit for a Linux-based Hadoop + cloud-native stack	Viable alternative if enterprise support is prioritized	Not selected — mixed OS estate not indicated

Storage comparison (Section E.3-d):

Criterion	SAN (selected)	NAS	Object Storage
Access pattern	Block-level, low-latency	File-level, shared access	REST/API, massively scalable
Best fit	High-performance VM/database storage	General file sharing	Long-term telemetry/archive storage
Role in this design	Primary storage for virtualized warehouse/tracking servers	Not primary — collaboration files only	Complementary — cold-tier archive for historical HDFS data

Trade-offs and justification: KVM was selected over VMware ESXi primarily on cost and native fit with the Linux-based Hadoop/cloud stack, accepting a smaller built-in management-tooling ecosystem in exchange for lower licensing cost — acceptable given the company's scale and cloud-native direction. SAN was selected as primary storage for its low-latency block access needed by transactional warehouse/VM workloads, with object storage used as a complementary cold tier rather than a SAN replacement, balancing performance against long-term storage cost. The route-optimization and predictive-maintenance results (Figures 3–4) validate that a MapReduce-style batch aggregation is sufficient for daily route summaries, while the clear separation of 3 flagged vehicles from 9 normal vehicles (Figure 4) confirms the chosen 25% anomaly-rate threshold gives a usable, low-noise maintenance signal; a lower threshold would increase false positives and technician workload, while a higher threshold risks missing early wear.

7. Broader Considerations

- Sustainability and environment — right-sized, elastic cloud/virtualized infrastructure avoids the energy waste of over-provisioned on-premises servers; route-optimization output directly reduces fuel consumption and emissions per shipment (SDG 9, SDG 12).
- Society and accessibility — a secure, standardized global-access layer (Section E.3-c) gives geographically distributed employees and partner logistics providers reliable access to the same platform, supporting inclusive, decent digital work (SDG 8) and quality technical upskilling for staff operating the new platform (SDG 4).
- Safety, security and ethics — predictive maintenance reduces the risk of in-transit vehicle failure (a road-safety benefit); sensor and location data are handled under the access-control and encryption measures in Section D to protect driver and partner privacy.
- Economics — cloud elasticity (Section E.2) converts capital expenditure on physical servers into operating expenditure aligned with actual usage, while fuel and maintenance savings from Figures 3–4 offset platform operating cost.
- Professional responsibility — the design documents explicit failure-handling and HA behavior (Figure 2) rather than assuming ideal conditions, reflecting responsible engineering practice for infrastructure the business depends on.

8. Conclusion

This assignment proposed and demonstrated an end-to-end cloud, virtualization and big-data architecture for a global logistics company's digital transformation: a KVM-based virtualization layer for warehouse and vehicle-tracking servers, a Zero-Trust/VPN global access layer, a SAN-backed storage tier, and a Hadoop-ecosystem big-data pipeline whose MapReduce design was implemented and executed as a local simulation. The simulation validated both required analytics outputs — route-level transit/fuel summaries across 6 routes and correct identification of 3 vehicles requiring predictive maintenance out of 12 — and confirmed that the design's failure-handling approach (speculative task execution, HDFS NameNode HA) prevents data loss on simulated component failure. Limitations: the demonstration used synthetic data at a reduced scale on a single-node local simulation rather than a live multi-node YARN/cloud cluster, so real-world network latency, multi-tenant contention, and actual cloud billing behavior were not measured. Possible improvements: deploy the same job logic on a managed Hadoop/Spark cluster (e.g., AWS EMR) against real telemetry, add a streaming (Kafka + Spark Streaming) path for near-real-time maintenance alerts rather than batch-only aggregation, and pilot the Zero-Trust access layer with a subset of partner logistics providers before global rollout.

9. Student Reflection

Beyond the taught theory, working through this assignment clarified how the individual course topics — service models, virtualization, SOA, and the Hadoop ecosystem — fit together as one pipeline rather than isolated concepts: the access and virtualization layers exist specifically to get trustworthy data into the big-data layer, and the big-data layer's value is only realized once its output (route/maintenance signals) is fed back into an operational decision. Actually implementing the MapReduce logic in code (rather than only diagramming it) surfaced a practical issue not obvious from theory alone — an unrealistic assumption of uniform vehicle wear produced no maintenance flags at all, which prompted a more realistic model with differentiated wear rates per vehicle. Given additional time and access to a live multi-node cluster, I would extend this into a streaming pipeline (Kafka + Spark Streaming) for near-real-time alerts, benchmark KVM against ESXi on real workloads rather than by criteria comparison alone, and validate the Zero-Trust access design against an actual partner integration. The exercise also reinforced the SDG linkages set out in Section A — the route-optimization signal is a direct, measurable route to SDG 9/12 fuel-and-emissions savings, and the secure global-access design is what makes SDG 8-aligned distributed work possible for this company in practice.

10. References

- P. Mell and T. Grance, "The NIST Definition of Cloud Computing," NIST Special Publication 800-145, National Institute of Standards and Technology, 2011.

- Apache Software Foundation, "Apache Hadoop Documentation" (HDFS Architecture, YARN, MapReduce Tutorial), hadoop.apache.org.
- Apache Software Foundation, "Apache Kafka Documentation," kafka.apache.org.
- T. White, Hadoop: The Definitive Guide, O'Reilly Media.
- Amazon Web Services, "AWS Well-Architected Framework" and "Amazon EMR Documentation," aws.amazon.com.
- VMware, "vSphere/ESXi Documentation"; Red Hat, "KVM/libvirt Documentation."
- NIST, "Zero Trust Architecture," NIST Special Publication 800-207, 2020.
- United Nations, "Sustainable Development Goals" (SDG 4, 8, 9, 12), sdgs.un.org.
- Python Software Foundation, matplotlib and Python 3 documentation (tools used for the local MapReduce simulation and result charts in Section E.5).
- AI-assisted tool disclosure: Claude (Anthropic) was used to help design, code, execute and document the local MapReduce simulation and accompanying figures in Section E.5, under the direction and review of the student.

F. Common Assessment Rubric

Assessment Criterion	Course-Specific Focus	Marks
Problem understanding & formulation	Understanding of the cloud migration and big-data problem, expected outcomes, assumptions and constraints for the logistics case	10
Application of course/domain knowledge	Application of cloud service models (IaaS/PaaS/SaaS/XaaS), virtualization, web services, SDDC/SOA and big-data (5Vs) concepts	20
Solution methodology / design / implementation	Virtualization architecture, hypervisor selection, SAN-based storage design, Hadoop/HDFS architecture and MapReduce job design	20
Use of appropriate modern tools / techniques	Evidence of tool usage such as Hadoop ecosystem, cloud platforms, virtualization/simulation tools, Git, MapReduce/YARN	10
Results, testing & validation	Presentation of outputs/screenshots, performance metrics and validation of the proposed architecture against stated requirements	15
Analysis, trade-offs & justification	Comparison of alternatives (hypervisors, storage, analytics techniques), security risk analysis and justification of final architecture	15
Broader considerations / professional responsibility, where applicable	Sustainability, ethics, security/privacy and societal impact of the proposed cloud and big-data solution (SDG 4, 8, 9, 12)	5
Technical documentation & reflection	Quality of documentation, references and student reflection on learning outcomes	5
TOTAL		100

F.1 Detailed Course-Specific Rubric

The detailed criterion-wise rubric below (aligned to CO1, CO2 and CO5) may be used by faculty for fine-grained evaluation within the marks allocated above.

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
1. Cloud Computing Evolution and Service Models (CO1)	5	Clearly analyzes the evolution of cloud computing from earlier computing models to modern cloud environments; thoroughly explains IaaS, PaaS, SaaS and XaaS and strongly justifies cloud adoption for the global logistics organization.	Explains cloud evolution and major service models correctly with minor gaps; provides reasonable justification for cloud adoption.	Provides a basic explanation of cloud computing evolution and service models, but analysis and organizational justification are limited.	Cloud concepts or evolution are poorly understood; service models are incorrectly explained or their relevance to logistics is unclear.

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
2. Server Virtualization and Web Services (CO1)	10	Clearly explains server virtualization, hypervisors, VMs, resource sharing, and web services; critically analyzes their role in enabling secure, scalable, and flexible cloud migration.	Correctly explains virtualization and web services with good connection to cloud migration; minor technical gaps exist.	Provides a basic explanation of virtualization and web services but gives limited analysis of their contribution to cloud migration.	Demonstrates poor understanding of virtualization or web services and fails to relate them effectively to cloud migration.
3. Virtualization Architecture, Hypervisor and Security (CO2)	10	Designs a clear virtualization architecture showing physical servers, suitable hypervisor, multiple VMs, storage, networking, and resource allocation; thoroughly analyzes virtualization security risks and appropriate mitigations.	Provides a suitable virtualization architecture and hypervisor selection with reasonable security analysis and minor gaps.	Shows a basic virtualization architecture and identifies some security risks, but resource allocation and mitigation analysis are limited.	Architecture is missing or technically incorrect; hypervisor selection and security risks/mitigations are poorly addressed.
4. Software Defined Datacenter, SOA and Cloud Federation (CO2)	5	Thoroughly analyzes SDDC and SOA and clearly demonstrates how software-defined compute, storage, networking, APIs, and service integration support global operations and cloud federation with logistics partners.	Explains SDDC and SOA correctly and describes cloud federation with minor omissions.	Provides a basic explanation of SDDC/SOA and mentions federation, but integration and organizational application are limited.	SDDC, SOA, or cloud federation is misunderstood or inadequately applied to the organization.
5. Global Cloud Access: Hardware, Client, Security and Network Requirements (CO2)	10	Provides a comprehensive analysis of client devices, hardware, network connectivity, bandwidth, authentication, authorization, encryption, VPN/zero-trust access, endpoint security, and secure global access for employees and partners.	Identifies major hardware, network, client, and security requirements with good justification and minor gaps.	Identifies basic access and network requirements but provides limited analysis of security, performance, or global connectivity.	Requirements are incomplete or inappropriate; secure cloud access and network considerations are poorly addressed.
6. SAN-Based Cloud Storage and Collaboration (CO2)	10	Designs a suitable SAN-based storage architecture with storage allocation, availability, scalability, backup/recovery, and data protection; clearly proposes collaboration standards and tools for globally distributed teams.	Provides a suitable SAN architecture and collaboration approach with minor technical gaps.	Describes basic SAN concepts and collaboration tools but provides limited analysis of scalability, availability, or recovery.	SAN architecture is unclear or incorrect; collaboration requirements and storage management are poorly addressed.

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
7. Distributed File Systems and 5Vs of Big Data (CO5)	10	Thoroughly analyzes distributed file-system requirements and clearly applies all 5Vs — Volume, Velocity, Variety, Veracity, and Value — to shipment, vehicle, warehouse, and IoT sensor data.	Correctly explains distributed file systems and the 5Vs with relevant logistics examples; minor omissions exist.	Provides a basic explanation of distributed storage and identifies some of the 5Vs, but application to the scenario is limited.	Demonstrates poor understanding of distributed file systems or the 5Vs; examples are missing or inappropriate.
8. Big Data Analytics, Integration and Security (CO5)	10	Proposes appropriate big-data analytics techniques and integration tools for shipment and sensor data; thoroughly addresses data ingestion, processing, visualization, security, access control, privacy, and governance.	Proposes suitable analytics and integration techniques with good security considerations and minor gaps.	Identifies basic analytics and integration approaches but provides limited justification or security analysis.	Analytics or integration techniques are inappropriate, poorly explained, or security considerations are largely absent.
9. Hadoop Architecture, HDFS and Ecosystem (CO5)	10	Designs and analyzes a comprehensive Hadoop architecture, including HDFS, NameNode, DataNodes, replication, fault tolerance, data movement, and relevant Hadoop ecosystem components; clearly relates the design to logistics data.	Explains Hadoop and HDFS architecture correctly with suitable ecosystem components and minor technical gaps.	Provides a basic description of Hadoop/HDFS and identifies some ecosystem components, but architecture and reliability analysis are limited.	Hadoop/HDFS concepts are poorly understood or the proposed architecture is technically incorrect/incomplete.
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Provides thoughtful justification of major design decisions, clearly connects the solution to relevant SDGs such as SDG 4, SDG 8, SDG 9 and SDG 12, and gives a specific account of challenges and learning gained.	Provides good justification of design decisions and identifies relevant SDGs; challenges and learning are mentioned with reasonable detail.	Reflection is present but generic; limited justification of design decisions, SDG relevance, or learning outcomes.	No meaningful reflection is provided, or the content lacks connection to the proposed architecture, SDGs, or learning outcomes.
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Provides thoughtful justification of major design decisions, clearly connects the solution to relevant SDGs such as SDG 4, SDG 8, SDG 9 and SDG 12, and gives a specific account of challenges and learning gained.	Provides good justification of design decisions and identifies relevant SDGs; challenges and learning are mentioned with reasonable detail.	Reflection is present but generic; limited justification of design decisions, SDG relevance, or learning outcomes.	No meaningful reflection is provided, or the content lacks connection to the proposed architecture, SDGs, or learning outcomes.

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Cloud evolution & service models (IaaS/PaaS/SaaS/XaaS)	CO1	PO1, PO2	L3/L4	5
Server virtualization & web services	CO1	PO1, PO2	L3/L4	10
Virtualization architecture, hypervisor & security	CO2	PO1, PO2, PO3	L4/L5	10
SDDC, SOA & cloud federation	CO2	PO2, PO3	L3/L4	5
Global cloud access: hardware, client, security & network	CO2	PO1, PO2	L4/L5	10
SAN-based cloud storage & collaboration	CO2	PO2, PO3	L4/L5	10
Distributed file systems & 5Vs of big data	CO5	PO1, PO2	L4/L5	10
Big data analytics, integration & security	CO5	PO2, PO3, PO5	L4/L5	10
Hadoop architecture, HDFS & ecosystem	CO5	PO1, PO2	L4/L5	10
MapReduce, YARN, failure handling & final architecture	CO5	PO2, PO3, PO5	L5/L6	10
Reflection: design decisions, SDG relevance & learning outcomes	CO1, CO2, CO5	PO6, PO7, PO12	L4/L5	10
TOTAL				100*

*Marks shown here follow the detailed course-specific rubric (F.1); faculty may normalize proportionally to the 100-mark scale in Section F where required. Faculty shall map only the POs and Bloom's levels genuinely assessed by this assignment.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60–69%
Level 1	50–59%
Level 0	< 50%

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well: _____

Areas requiring improvement: _____

Corrective / Improvement Action: _____

Action to be implemented in the next offering: _____

Documents to be Maintained

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports
7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action